PART - A: MULTIPLE CHOICE QUESTIONS.

(Answer all Questions. Each question carries **1 Mark)**

1. The component form of the vector with initial point $A=(2,3,4)$ and terminal point $B=(5,1,2)$ is :

   a) $\overrightarrow{AB}=(-3,-2,2)$ b) $\overrightarrow{AB}=(-3,2,2)$ c) $\overrightarrow{AB}=(3,-2,2)$ d) $\overrightarrow{AB}=(3,-2,-2)$

2. If $\vec{i}$, $\vec{j}$ and $\vec{k}$ are the standard unit vectors in the space, then $2\cdot\vec{j}-3\cdot\vec{k}=$_____.

   a) $(2,2,-3)$ b) $(0,2,-3)$ c) $(2,-3,0)$ d) $(0,-2,3)$

3. If $|A|=6$, then $|A^{-1}|=$_____.

   a) 6 b) -6 c) 36 d) $\frac{1}{6}$

4. If $A=\{2,4,5,6,7\}$ and $B=\{1,2,3,6,9\}$, then $A\oplus B=$_____.

   a) $\{1,3,4,5,7,9\}$ b) ϕ c) $\{2,6\}$ d) $\{1,2,3,4,5,6,7,9\}$

5. How many different number plates are possible with a lead digit of 4 or 5 or 6 followed by 2 letters and then 2 digits?

   a) 175500 b) 67600 c) 202800 d) 58500

6. When two coins are tossed at a time, what is the probability of getting at least one Head?

   a) $\frac{3}{4}$ b) $\frac{1}{4}$ c) $\frac{1}{2}$ d) 1

7. If A and B are independent events, then $P(A^c\cap B)=$_____ .

   a) $P(A)\cdot P(B^c)$ b) $P(A^c)\cdot P(B)$ c) $P(A)\cdot P(B)$ d) $P(A^c)\cdot P(B^c)$

8. Which of the following numbers are relatively prime?

   a) $11,16$ b) $25,15$ c) $6,15$ d) $12,16$

9. Find the value of b from the following which satisfies $24\equiv b \pmod 4$.

   a) 5 b) 1 c) 4 d) 2

10. When -67 is divided by 23, what are the quotient and remainder ?

    a) 2 and -3 b) -3 and 2 c) -2 and -21 d) -21 and 2

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PART - B: SHORT ANSWER QUESTIONS

(Answer all questions. Each question carries **3 Marks**)

1. Find the angle (in degrees) between the vectors $\vec{u}=(2,-1,6)$ and $\vec{v}=(-2,3,5)$.

Solution:

Now, $|\vec{u}|=\sqrt{(2)^2+(-1)^2+(6)^2}=\sqrt{41}$, $|\vec{v}|=\sqrt{(-2)^2+(3)^2+(5)^2}=\sqrt{38}$ **[1 M]**

and $\vec{u}\cdot\vec{v}=(2)\cdot(-2)+(-1)\cdot(3)+(6)\cdot(5)=23$ **[0.5 M]**

Let θ be the angle between the vectors $\vec{u}=(2,-1,6)$ and $\vec{v}=(-2,3,5)$.

Then, $\theta=\cos^{-1}\left(\dfrac{\vec{u}\cdot\vec{v}}{|\vec{u}||\vec{v}|}\right)=\cos^{-1}\left(\dfrac{23}{\sqrt{41}\sqrt{38}}\right)=\cos^{-1}(0.5827)=54.36^0$ **[1.5 M]**

2. Find the values of x, y and z, from the given equal matrices. $\begin{bmatrix} x+y & 2 \\ 5+z & xy \end{bmatrix}=\begin{bmatrix} 6 & 2 \\ 5 & 8 \end{bmatrix}$.

Solution:

From the given equal matrices, we get

$x+y=6----(1)$
$5+z=5----(2)$ **[1 M]**
$xy=8-----(3)$

From eq (2), we get

$5+z=5$
$z=0$

From eq (3) , we get

$xy=8$
$x=\dfrac{8}{y}$ **[0.5+0.5 M]**

Substituting in eq (1)

$\dfrac{8}{y}+y=6$
$y^2-6y+8=0$. **[1 M]**
$\Rightarrow y=4,\ y=2$
then, $x=2, x=4$

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3. A group consists of 3 girls and 5 boys. A team of 4 members to be selected from the group.
   (i) Find the number of teams having equal number of boys and girls. **[1 M]**
   (ii) Find the number of teams having exactly one boy. **[1 M]**
   (iii) Find the number of teams having at most one girl. **[1 M]**

Solution: Total number of students in the group are: 3+5 = 8. We have to select 4 students team from 8 students.

(i) A team having equal number of boys and girls means, we have to select 2 boys from 5 boys and 2 girls from 3 girls. Therefore, the number of teams having equal number of boys and girls are : $5_{C_2} \cdot 3_{C_2} = 10 \cdot 3 = 30$. **[1 M]**

(ii) A team having exactly one boy means, we have to select 1 boy from 5 boys and 3 girls from 3 girls. Therefore, the number of teams having exactly one boy are: $5_{C_1} \cdot 3_{C_3} = 5 \cdot 1 = 5$. **[1 M]**

(iii) A team having at most one girl means, we have two choices (1 girl and 3 boys or no girl and 4 boys). Therefore, the number of teams having at most one girl are: $3_{C_1} \cdot 5_{C_3} + 3_{C_0} \cdot 5_{C_4} = 3 \cdot 10 + 1 \cdot 5 = 35$. **[1 M]**

4. A family has three children. Find the conditional probability of the family that has two boys, provided that the family has at least one boy.

(OR)

A box contains 4 red and 6 white balls. One by one 3 balls are drawn at random without replacement.
   (i) What is the probability that all are same color? **[1.5 M]**
   (ii) What is the probability that the color of the second is opposite to the first and third? **[1.5 M]**

Solution:

If a family has 3 children, then sample space $S = \{BBB, BBG, BGB, GBB, BGG, GBG, GGB, GGG\}$.

Then $n(S) = 8$.

Let $A =$ a family has two boys $= \{BBG, BGB, GBB\} \Rightarrow n(A) = 3$,

$B =$ a family has at least one boy $= \{BBB, BBG, BGB, GBB, BGG, GBG, GGB\} \Rightarrow n(B) = 7$,

and $A \cap B = \{BBG, BGB, GBB\} \Rightarrow n(A \cap B) = 3$. **[1 M]**

Now, $P(A) = \dfrac{n(A)}{n(S)} = \dfrac{3}{8}$, $P(\text{B}) = \dfrac{n(B)}{n(S)} = \dfrac{7}{8}$, and $P(A \cap B) = \dfrac{n(A \cap B)}{n(S)} = \dfrac{3}{8}$. **[1 M]**

Therefore, the conditional probability of the family that has two boys, provided that the family has at least one boy : $P(A/B) = \dfrac{P(A \cap B)}{P(B)} = \dfrac{(3/8)}{(7/8)} = \dfrac{3}{7} = 0.429$. **[1 M]**

(OR)

Here the total number of balls are: 4 + 6 =10. One by one 3 balls are drawn at random without replacement.

(i) If all balls are same color, then we have two choices (either all are red or all are white).

Therefore, the probability that all are same color:

$$P(RRR\ or\ WWW) = P(RRR) + P(WWW)$$
$$= P(R)\cdot P(R)\cdot P(R) + P(W)\cdot P(W)\cdot P(W)$$ **[1.5 M]**
$$= \frac{4}{10}\cdot\frac{3}{9}\cdot\frac{2}{8} + \frac{6}{10}\cdot\frac{5}{9}\cdot\frac{4}{8} = \frac{24}{720} + \frac{120}{720} = \frac{144}{720} = 0.194$$

(ii) If the color of the second is opposite to the first and third, then we have two choices (first and third are red and second is white or first and third are white and second is red)

Therefore, the probability that the color of the second is opposite to the first and third:

$$P(RWR\ or\ WRW) = P(RWR) + P(WRW)$$
$$= P(R)\cdot P(W)\cdot P(R) + P(W)\cdot P(R)\cdot P(W)$$ **[1.5 M]**
$$= \frac{4}{10}\cdot\frac{6}{9}\cdot\frac{3}{8} + \frac{6}{10}\cdot\frac{4}{9}\cdot\frac{5}{8} = \frac{72}{720} + \frac{120}{720} = \frac{192}{720} = 0.267$$

5. Decrypt the message DEZA ESP MFD encrypted by using the shift cipher $f(p) = (p+11) \bmod 26$.

Solution: Given encrypted message is: DEZA ESP MFD

To decrypt the message, we use $f^{-1}(c) = (c-11) \bmod 26$ as a decryption function.

Step – (1): Replace the letters in the message with the corresponding numbers (element of Z_{26}), we get

D	E	Z	A	E	S	P	M	F	D
3	4	25	0	4	18	15	12	5	3

[1 M]

Step – (2): Now replace each of these numbers ‘c’ by using $f^{-1}(c) = (c-11) \bmod 26$, we get

3	4	25	0	4	18	15	12	5	3
18	19	14	15	19	7	4	1	20	18

[1 M]

Step – (3): Translating this back to letters, we get

18	19	14	15	19	7	4	1	20	18
S	T	O	P	T	H	E	B	U	S

[1 M]

The decrypted message is: STOP THE BUS

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PART - C: LONG ANSWER QUESTIONS

(Answer all questions. Each question carries **5 Marks**)

1. Use Cramer's rule to solve the following system of linear equations:

$x+2y+z=7,\quad 2x-y+2z=4,\quad x+y-2z=-1.$

Solution:

Given system of linear equations : $\left.\begin{array}{l} x+2y+z=7, \\ 2x-y+2z=4, \\ x+y-2z=-1 \end{array}\right\} ------ (1)$

The matrix form of (1) is : $AX=B$, where $A=\begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 1 & 1 & -2 \end{bmatrix}$, $X=\begin{bmatrix} x \\ y \\ z \end{bmatrix}$, $B=\begin{bmatrix} 7 \\ 4 \\ -1 \end{bmatrix}$. **[0.5 M]**

Now, $|A|=\begin{vmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 1 & 1 & -2 \end{vmatrix}=1\cdot(2-2)-2\cdot(-4-2)+1\cdot(2+1)=15\neq 0$. **[1 M]**

So we can apply Cramer's rule to find the solution of (1).

Now,

$|A_1|=\begin{vmatrix} 7 & 2 & 1 \\ 4 & -1 & 2 \\ -1 & 1 & -2 \end{vmatrix}=7\cdot(2-2)-2\cdot(-8+2)+1\cdot(4-1)=15,$ **[1 M]**

$|A_2|=\begin{vmatrix} 1 & 7 & 1 \\ 2 & 4 & 2 \\ 1 & -1 & -2 \end{vmatrix}=1\cdot(-8+2)-7\cdot(-4-2)+1\cdot(-2-4)=30,$ **[1 M]**

$|A_3|=\begin{vmatrix} 1 & 2 & 7 \\ 2 & -1 & 4 \\ 1 & 1 & -1 \end{vmatrix}=1\cdot(1-4)-2\cdot(-2-4)+7\cdot(2+1)=30.$ **[1 M]**

Finally, $x=\dfrac{|A_1|}{|A|}=\dfrac{15}{15}=1,\ y=\dfrac{|A_2|}{|A|}=\dfrac{30}{15}=2,\ z=\dfrac{|A_3|}{|A|}=\dfrac{30}{15}=2.$ **[0.5 M]**

Therefore, the solution of (1) is : $x=1,\ y=2\ \ z=2$.

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2. Compute the following sums:

(i) $\sum_{k=1}^{5}(2k+3)$ **[2.5 M]**

(ii) $\sum_{i=0}^{2}\sum_{j=0}^{1}(3i+4j)$ **[2.5 M]**

Solution:

(i) Now

$$\begin{aligned}\sum_{k=1}^{5}(2k+3) &= \sum_{k=1}^{5}(2k)+\sum_{k=1}^{5}(3)\\ &= 2\cdot\sum_{k=1}^{5}(k)+3\cdot\sum_{k=1}^{5}(1)\\ &= 2\cdot\frac{5\cdot(5+1)}{2}+3\cdot 5 \qquad \textbf{[2.5 M]}\\ &= 30+15\\ &= 45\end{aligned}$$

(ii) Now

$$\begin{aligned}\sum_{i=0}^{2}\sum_{j=0}^{1}(3i+4j) &= \sum_{i=0}^{2}\left[(3i+4\cdot 0)+(3i+4\cdot 1)\right]\\ &= \sum_{i=0}^{2}(6i+4)\\ &= (6\cdot 0+4)+(6\cdot 1+4)+(6\cdot 2+4) \qquad \textbf{[2.5 M]}\\ &= 4+10+16\\ &= 30\end{aligned}$$

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3. Four cards are drawn at a time from a deck of 52 cards.

(i) What are the number of ways of choosing 4 cards? **[1 M]**

(ii) What are the number of ways of choosing 4 cards coming from four different suits? **[2 M]**

(iii) What are the number of ways of choosing 4 cards coming from the same suit? **[2 M]**

(OR)

(a) Find the number of 4-digit numbers that can be formed using the digits 2, 3, 0, 6, 8 if no digit is repeated. How many of these 4-digit numbers will be odd? **[2 M]**

(b) Find the number of different 8-letter arrangements that can be made from the letters of the word REPEATED. In how many of these arrangements do all the vowels occur together? **[3 M]**

Solution:

(i) The number of ways choosing 4 cards from 52 cards are : $52_{C_4} = 270725$ **[1 M]**

(ii) In 52 cards, there are four different suits namely, diamond, heart club, and spade. Also there are 13 cards of each suit. In this case we can select one card from each suit.

Therefore, the number of ways of choosing 4 cards coming from four different suits are :

$13_{C_1} \cdot 13_{C_1} \cdot 13_{C_1} \cdot 13_{C_1} = 13^4 = 28561$. **[2 M]**

(iii) In this case we can select four cards from each suit. Then we have 4 choices.
Therefore, the number of ways of choosing 4 cards coming from the same suit are :

$13_{C_4} + 13_{C_4} + 13_{C_4} + 13_{C_4} = 715 + 715 + 715 + 715 = 2860$. **[2 M]**

(OR)

(a) The number of 4-digit numbers that can be formed using the digits 2, 3, 0, 6, 8 if no digit is repeated are : $4 \cdot 4 \cdot 3 \cdot 2 = 96$ (first place cannot be filled with digit 0) **[1 M]**

Also, the number of 4-digit odd numbers that can be formed using the digits 2, 3, 0, 6, 8 are :

$3 \cdot 3 \cdot 2 \cdot 1 = 18$ (first place cannot be filled with 0 and last place can be filled with only 3) **[1 M]**

(b) The word REPEATED contains 8 letters including 4 vowels E, E, E, A.
The number of different 8-letter arrangements that can be made from the letters of the word REPEATED are : $\frac{8!}{3!} = 6720$. **[1 M]**

Since the vowels have to occur together, we can assume all the vowels E, E, E, A as a single object. This single object together with 4 remaining letters (R, P, T, D) will be counted as 5 objects. Now the number of arrangements with the 5 objects would be : $5_{P_5} = 5! = 120$. **[1 M]**

Corresponding to each of these 120 arrangements, we shall have $\frac{4!}{3!} = 4$ arrangements of the four vowels E, E, E, A. Therefore, the number of different 8-letter arrangements in which vowels occur together are : $5_{P_5} \cdot \frac{4!}{3!} = 120 \cdot 4 = 480$ **[1 M]**

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4. Consider a random variable X with the following probability distribution:

$X = x$	1	2	3	4	5
$P(X = x)$	0.2	0.1	K	0.3	0.2

(i) Find the value of K. **[1 M]**

(ii) What is the expectation of X ? **[1 M]**

(iii) Evaluate the variance and standard deviation of X. **[2 M]**

(iv) Find $P(1 < X \leq 5)$. **[1 M]**

Solution:

(i) By the property of Probability distribution:

$$\sum_{x=1}^{5} P(X = x) = 1 \Rightarrow 0.2 + 0.1 + \text{K} + 0.3 + 0.2 = 1$$

$$\Rightarrow K = 0.2$$ **[1 M]**

(ii) Now, expectation of X :

$$E(X) = \mu = \sum_{x=1}^{5} x \cdot P(X = x)$$

$$= (1)\cdot(0.2) + (2)\cdot(0.1) + (3)\cdot(0.2) + (4)\cdot(0.3) + (5)\cdot(0.2)$$ **[1 M]**

$$= 3.2$$

(iii) Now, variance of X :

$$V(X) = \sigma^2 = E(X^2) - \left[E(X)\right]^2$$

$$= \sum_{x=1}^{5} x^2 \cdot P(X = x) - \left[\mu\right]^2$$

$$= \sum_{x=1}^{5} \left[(1^2)\cdot(0.2) + (2^2)\cdot(0.1) + (3^2)\cdot(0.2) + (4^2)\cdot(0.3) + (5^2)\cdot(0.2)\right] - \left[3.2\right]^2$$ **[1.5 M]**

$$= 12.2 - 10.24$$

$$= 1.96$$

and Standard deviation of X: $\sigma = \sqrt{\sigma^2} = \sqrt{1.96} = 1.4$ **[0.5 M]**

(iv) Now,

$$P(1 < X \leq 5) = P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5)$$

$$= 0.1 + 0.2 + 0.3 + 0.2$$ **[1 M]**

$$= 0.8$$

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5. Solve the linear congruence: $26x \equiv 1 \pmod{49}$

Solution:

Step – 1: Using Euclidean Algorithm, we show that $\gcd(26, 49) = 1$.

Now, using Division Algorithm:

$$49 = 26 \cdot 1 + 23 \Rightarrow 49 + 26 \cdot (-1) = 23 \text{ --------- } (1)$$
$$26 = 23 \cdot 1 + 3 \Rightarrow 26 + 23 \cdot (-1) = 3 \text{ --------- } (2)$$
$$23 = 3 \cdot 7 + 2 \Rightarrow 23 + 3 \cdot (-7) = 2 \text{ --------- } (3)$$ **[2 M]**
$$3 = 2 \cdot 1 + 1 \Rightarrow 3 + 2 \cdot (-1) = 1 \text{ --------- } (4)$$
$$2 = 1 \cdot 2 + 0$$

Therefore, inverse of 26 under modulo 49 exists, since $\gcd(26, 49) = 1$ (non-zero remainder). **[0.5 M]**

Step – 2: We can express 1 as a linear combination of 26 and 49.

Now

$$\begin{aligned}
(4) &\Rightarrow 3 + 2 \cdot (-1) = 1 \\
&\Rightarrow 3 + [23 + 3 \cdot (-7)] \cdot (-1) = 1, \text{ from (3)} \\
&\Rightarrow 3 \cdot 1 + 23 \cdot (-1) + 3 \cdot 7 = 1 \\
&\Rightarrow 3 \cdot 8 + 23 \cdot (-1) = 1 \\
&\Rightarrow [26 + 23 \cdot (-1)] \cdot 8 + 23 \cdot (-1) = 1, \text{ from (2)} \\
&\Rightarrow 26 \cdot 8 + 23 \cdot (-8) + 23 \cdot (-1) = 1 \\
&\Rightarrow 26 \cdot 8 + 23 \cdot (-9) = 1 \\
&\Rightarrow 26 \cdot 8 + [49 + 26 \cdot (-1)] \cdot (-9) = 1, \text{ from (1)} \\
&\Rightarrow 26 \cdot 8 + 49 \cdot (-9) + 26 \cdot 9 = 1 \\
&\Rightarrow 26 \cdot 17 + 49 \cdot (-9) = 1
\end{aligned}$$

[2 M]

Taking modulo 49 on both sides, we get : $26 \cdot 17 \equiv 1 \pmod{49}$.

Therefore, the solution of the given linear congruence is : $x = 17$. **[0.5 M]**

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PART - A: MULTIPLE CHOICE QUESTIONS.

(Answer all Questions. Each question carries **1 Mark)**

1. The component form of the vector with initial point $A=(5,1,2)$ and terminal point $B=(2,3,4)$ is :

   a) $\overrightarrow{AB}=(-3,-2,2)$ b) $\overrightarrow{AB}=(-3,2,2)$ c) $\overrightarrow{AB}=(3,-2,2)$ d) $\overrightarrow{AB}=(3,-2,-2)$

2. If $\vec{i}$, $\vec{j}$ and $\vec{k}$ are the standard unit vectors in the space, then $4\cdot\vec{j}-5\cdot\vec{k}=$_____

   a) $(1,4,-5)$ b) $(4,-5,0)$ c) $(0,4,-5)$ d) $(0,-4,5)$

3. If $|A|=8$, then $|A^{-1}|=$_____ .

   a) 8 b) $\frac{1}{8}$ c) 64 d) -8

4. If $A=\{0,4,5,7,8\}$ and $B=\{1,3,5,6,8\}$, then $A\oplus B=$_____

   a) $\{5,8\}$ b) ϕ c) $\{0,1,3,4,5,6,7,8\}$ d) $\{0,1,3,4,6,7\}$

5. How many different number plates are possible with a lead digit of 3 or 5 or 7 followed by 2 letters and then 2 digits?

   a) 202800 b) 67600 c) 175500 d) 58500

6. When two coins are tossed at a time, what is the probability of getting at least one Tail?

   a) 1 b) $\frac{1}{4}$ c) $\frac{1}{2}$ d) $\frac{3}{4}$

7. If A and B are independent events, then $P(A\cap B^c)=$_____

   a) $P(A)\cdot P(B^c)$ b) $P(A^c)\cdot P(B)$ c) $P(A)\cdot P(B)$ d) $P(A^c)\cdot P(B^c)$

8. Which of the following numbers are relatively prime?

   a) 20, 15 b) 13, 16 c) 6, 12 d) 12, 18

9. Find the value of b from the following which satisfies $28\equiv b \pmod 4$.

   a) 5 b) 1 c) 2 d) 4

10. When -61 is divided by 21, what are the quotient and remainder ?

    a) 2 and -3 b) -2 and -15 c) -3 and 2 d) -15 and 2

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PART - B: SHORT ANSWER QUESTIONS

(Answer all questions. Each question carries **3 Marks**)

1. Find the angle (in degrees) between the vectors $\vec{u}=(2,-1,5)$ and $\vec{v}=(-2,3,4)$.

Solution:

Now, $|\vec{u}|=\sqrt{(2)^2+(-1)^2+(5)^2}=\sqrt{30}$, $|\vec{v}|=\sqrt{(-2)^2+(3)^2+(4)^2}=\sqrt{29}$ **[1 M]**

and $\vec{u}\cdot\vec{v}=(2)\cdot(-2)+(-1)\cdot(3)+(5)\cdot(4)=13$ **[0.5 M]**

Let θ be the angle between the vectors $\vec{u}=(2,-1,5)$ and $\vec{v}=(-2,3,4)$.

Then, $\theta=\cos^{-1}\left(\dfrac{\vec{u}\cdot\vec{v}}{|\vec{u}||\vec{v}|}\right)=\cos^{-1}\left(\dfrac{13}{\sqrt{30}\sqrt{29}}\right)=\cos^{-1}(0.4407)=63.85^0$ **[1.5 M]**

2. Find the values of x, y and z, from the given equal matrices. $\begin{bmatrix} x+y & 2 \\ 5+z & xy \end{bmatrix}=\begin{bmatrix} 8 & 2 \\ 5 & 12 \end{bmatrix}$.

Solution:

From the given equal matrices, we get

$x+y=8----(1)$
$5+z=5----(2)$ **[1 M]**
$xy=12-----(3)$

From eq (2), we get

$5+z=5$
$z=0$

From eq (3) , we get

$xy=12$
$x=\dfrac{12}{y}$ **[0.5+0.5 M]**

Substituting in eq (1)

$\dfrac{12}{y}+y=8$
$y^2-8y+12=0$ **[1 M]**
$\Rightarrow y=6,\ y=2$
then, $x=2, x=6$

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3. A group consists of 5 girls and 3 boys. A team of 4 members to be selected from the group.

(i) Find the number of teams having equal number of girls and boys. **[1 M]**

(ii) Find the number of teams having exactly one girl. **[1 M]**

(iii) Find the number of teams having at most one boy. **[1 M]**

Solution: Total number of students in the group are: 5+3 = 8. We have to select 4 students team from 8 students.

(i) A team having equal number of girls and boys means, we have to select 2 girls from 5 girls and 2 boys from 3 boys. Therefore, the number of teams having equal number of boys and girls are : $5_{C_2} \cdot 3_{C_2} = 10 \cdot 3 = 30$. **[1 M]**

(ii) A team having exactly one girl means, we have to select 1 girl from 5 girls and 3 boys from 3 boys. Therefore, the number of teams having exactly one girl are: $5_{C_1} \cdot 3_{C_3} = 5 \cdot 1 = 5$. **[1 M]**

(iii) A team having at most one boy means, we have two choices (1 boy and 3 girls or no boy and 4 girls). Therefore, the number of teams having at most one boy are: $3_{C_1} \cdot 5_{C_3} + 3_{C_0} \cdot 5_{C_4} = 3 \cdot 10 + 1 \cdot 5 = 35$. **[1 M]**

4. A family has three children. Find the conditional probability of the family that has two girls, provided that the family has at least one girl.

(OR)

A box contains 6 red and 4 white balls. One by one 3 balls are drawn at random without replacement.

(i) What is the probability that all are same color? **[1.5 M]**

(ii) What is the probability that the color of the second is opposite to the first and third? **[1.5 M]**

Solution:

If a family has 3 children, then sample space $S = \{BBB, BBG, BGB, GBB, BGG, GBG, GGB, GGG\}$.

Then $n(S) = 8$.

Let $A =$ a family has two girls $= \{BGG, GBG, GGB\} \Rightarrow n(A) = 3$,

$B =$ a family has at least one girl $= \{BBG, BGB, GBB, BGG, GBG, GGB, GGG\} \Rightarrow n(B) = 7$,

and $A \cap B = \{BGG, GBG, GGB\} \Rightarrow n(A \cap B) = 3$. **[1 M]**

Now, $P(A) = \dfrac{n(A)}{n(S)} = \dfrac{3}{8}$, $P(\mathrm{B}) = \dfrac{n(B)}{n(S)} = \dfrac{7}{8}$, and $P(A \cap B) = \dfrac{n(A \cap B)}{n(S)} = \dfrac{3}{8}$. **[1 M]**

Therefore, the conditional probability of the family that has two girls, provided that the family has at least one girl : $P(A/B) = \dfrac{P(A \cap B)}{P(B)} = \dfrac{(3/8)}{(7/8)} = \dfrac{3}{7} = 0.429$. **[1 M]**

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(OR)

Here the total number of balls are: 6 + 4 =10. One by one 3 balls are drawn at random without replacement.

(i) If all balls are same color, then we have two choices (either all are red or all are white).

Therefore, the probability that all are same color:

$$P(RRR\ or\ WWW) = P(RRR) + P(WWW)$$

$$= P(R)\cdot P(R)\cdot P(R) + P(W)\cdot P(W)\cdot P(W)$$ **[1.5 M]**

$$= \frac{6}{10}\cdot\frac{5}{9}\cdot\frac{4}{8} + \frac{4}{10}\cdot\frac{3}{9}\cdot\frac{2}{8} = \frac{120}{720} + \frac{24}{720} = \frac{144}{720} = 0.194$$

(ii) If the color of the second is opposite to the first and third, then we have two choices (first and third are red and second is white or first and third are white and second is red)

Therefore, the probability that the color of the second is opposite to the first and third:

$$P(RWR\ or\ WRW) = P(RWR) + P(WRW)$$

$$= P(R)\cdot P(W)\cdot P(R) + P(W)\cdot P(R)\cdot P(W)$$ **[1.5 M]**

$$= \frac{6}{10}\cdot\frac{4}{9}\cdot\frac{5}{8} + \frac{4}{10}\cdot\frac{6}{9}\cdot\frac{3}{8} = \frac{120}{720} + \frac{72}{720} = \frac{192}{720} = 0.267$$

5. Decrypt the message DPYO STX ZFE encrypted by using the shift cipher $f(p) = (p+11) \bmod 26$.

Solution: Given encrypted message is: DPYO STX ZFE

To decrypt the message, we use $f^{-1}(\text{c}) = (\text{c}-11) \bmod 26$ as a decryption function.

Step – (1): Replace the letters in the message with the corresponding numbers (element of Z_{26}), we get

D	P	Y	O	S	T	X	Z	F	E
3	15	24	14	18	19	23	25	5	4

[1 M]

Step – (2): Now replace each of these numbers ‘c’ by using $f^{-1}(\text{c}) = (\text{c}-11) \bmod 26$, we get

3	15	24	14	18	19	23	25	5	4
18	4	13	3	7	8	12	14	20	19

[1 M]

Step – (3): Translating this back to letters, we get

18	4	13	3	7	8	12	14	20	19
S	E	N	D	H	I	M	O	U	T

[1 M]

The decrypted message is: SEND HIM OUT

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PART - C: LONG ANSWER QUESTIONS

(Answer all questions. Each question carries **5 Marks**)

1. Use Cramer's rule to solve the following system of linear equations :

$3x+y-z=2,\quad 2x-y+2z=6,\quad 2x+y-2z=-2$.

Solution:

Given system of linear equations : $\left.\begin{array}{l}3x+y-z=2,\\ 2x-y+2z=6,\\ 2x+y-2z=-2\end{array}\right\}$ ------ (1)

The matrix form of (1) is : $AX=B$, where $A=\begin{bmatrix}3 & 1 & -1\\ 2 & -1 & 2\\ 2 & 1 & -2\end{bmatrix}$, $X=\begin{bmatrix}x\\ y\\ z\end{bmatrix}$, $B=\begin{bmatrix}2\\ 6\\ -2\end{bmatrix}$. **[0.5 M]**

Now, $|A|=\begin{vmatrix}3 & 1 & -1\\ 2 & -1 & 2\\ 2 & 1 & -2\end{vmatrix}=3\cdot(2-2)-1\cdot(-4-4)-1\cdot(2+2)=4\neq 0$. **[1 M]**

So we can apply Cramer's rule to find the solution of (1).

Now,

$|A_1|=\begin{vmatrix}2 & 1 & -1\\ 6 & -1 & 2\\ -2 & 1 & -2\end{vmatrix}=2\cdot(2-2)-1\cdot(-12+4)-1\cdot(6-2)=4,$ **[1 M]**

$|A_2|=\begin{vmatrix}3 & 2 & -1\\ 2 & 6 & 2\\ 2 & -2 & -2\end{vmatrix}=3\cdot(-12+4)-2\cdot(-4-4)-1\cdot(-4-12)=8,$ **[1 M]**

$|A_3|=\begin{vmatrix}3 & 1 & 2\\ 2 & -1 & 6\\ 2 & 1 & -2\end{vmatrix}=3\cdot(2-6)-1\cdot(-4-12)+2\cdot(2+2)=12.$ **[1 M]**

Finally, $x=\dfrac{|A_1|}{|A|}=\dfrac{4}{4}=1,\ y=\dfrac{|A_2|}{|A|}=\dfrac{8}{4}=2,\ z=\dfrac{|A_3|}{|A|}=\dfrac{12}{4}=3$. **[0.5 M]**

Therefore, the solution of (1) is : $x=1,\ y=2,\ z=3$.

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2. Compute the following sums:

(i) $\sum_{k=1}^{5}(3k+2)$ **[2.5 M]**

(ii) $\sum_{i=0}^{2}\sum_{j=0}^{1}(4i+3j)$ **[2.5 M]**

Solution:

(i) Now

$$\begin{aligned}\sum_{k=1}^{5}(3k+2) &= \sum_{k=1}^{5}(3k)+\sum_{k=1}^{5}(2)\\ &= 3\cdot\sum_{k=1}^{5}(k)+2\cdot\sum_{k=1}^{5}(1)\\ &= 3\cdot\frac{5\cdot(5+1)}{2}+2\cdot 5 \qquad \textbf{[2.5 M]}\\ &= 45+10\\ &= 55\end{aligned}$$

(ii) Now

$$\begin{aligned}\sum_{i=0}^{2}\sum_{j=0}^{1}(4i+3j) &= \sum_{i=0}^{2}\left[(4i+3\cdot 0)+(4i+3\cdot 1)\right]\\ &= \sum_{i=0}^{2}(8i+3)\\ &= (8\cdot 0+3)+(8\cdot 1+3)+(8\cdot 2+3) \qquad \textbf{[2.5 M]}\\ &= 3+11+19\\ &= 33\end{aligned}$$

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3. Four cards are drawn at a time from a deck of 52 cards.

(i) What are the number of ways of choosing 4 cards? **[1 M]**

(ii) What are the number of ways of choosing 2 red cards and 2 black cards? **[2 M]**

(iii) What are the number of ways of choosing 4 cards from the same color? **[2 M]**

(OR)

(a) Find the number of 4-digit numbers that can be formed using the digits 1, 2, 0, 4, 6 if no digit is repeated. How many of these 4-digit numbers will be odd? **[2 M]**

(b) Find the number of different 8-letter arrangements that can be made from the letters of the word VARADALU. In how many of these arrangements do all the vowels occur together? **[3 M]**

Solution:

(i) The number of ways choosing 4 cards from 52 cards are : $52_{C_4} = 270725$. **[1 M]**

(ii) In 52 cards, there are 26 red cards and 26 black cards.

Therefore, the number of ways of choosing 2 red cards and 2 black cards are :

$26_{C_2} \cdot 26_{C_2} = 325^2 = 105625$. **[2 M]**

(iii) In this case we can select four cards from each color. Then we have 2 choices.

Therefore, the number of ways of choosing 4 cards from the same color are :

$26_{C_4} + 26_{C_4} = 14950 + 14950 = 29900$. **[2 M]**

(OR)

(a) The number of 4-digit numbers that can be formed using the digits 1, 2, 0, 4, 6 if no digit is repeated are : $4 \cdot 4 \cdot 3 \cdot 2 = 96$ (first place cannot be filled with digit 0) **[1 M]**

Also, the number of 4-digit odd numbers that can be formed using the digits 1, 2, 0, 4, 6 are :

$3 \cdot 3 \cdot 2 \cdot 1 = 18$ (first place cannot be filled with 0 and last place can be filled with only 1) **[1 M]**

(b) The word VARADALU contains 8 letters including 4 vowels A, A, A, U.

The number of different 8-letter arrangements that can be made from the letters of the word VARADALU are : $\frac{8!}{3!} = 6720$. **[1 M]**

Since the vowels have to occur together, we can assume all the vowels A, A, A, U as a single object. This single object together with 4 remaining letters (V, R, D, L) will be counted as 5 objects. Now the number of arrangements with the 5 objects would be : $5_{P_5} = 5! = 120$. **[1 M]**

Corresponding to each of these 120 arrangements, we shall have $\frac{4!}{3!} = 4$ arrangements of the four vowels A, A, A, U. Therefore, the number of different 8-letter arrangements in which vowels occur together are : $5_{P_5} \cdot \frac{4!}{3!} = 120 \cdot 4 = 480$ **[1 M]**

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4. Consider a random variable X with the following probability distribution:

$X = x$	1	2	3	4	5
$P(X = x)$	0.1	0.2	K	0.2	0.3

(i) Find the value of K. **[1 M]**

(ii) What is the expectation of X ? **[1 M]**

(iii) Evaluate the variance and standard deviation of X . **[2 M]**

(iv) Find $P(1 < X \le 5)$. **[1 M]**

Solution:

(i) By the property of Probability distribution:

$$\sum_{x=1}^{5} P(X = x) = 1 \Rightarrow 0.1 + 0.2 + \text{K} + 0.2 + 0.3 = 1$$

$$\Rightarrow K = 0.2$$ **[1 M]**

(ii) Now, expectation of X :

$$E(X) = \mu = \sum_{x=1}^{5} x \cdot P(X = x)$$

$$= (1)\cdot(0.1) + (2)\cdot(0.2) + (3)\cdot(0.2) + (4)\cdot(0.2) + (5)\cdot(0.3)$$ **[1 M]**

$$= 3.4$$

(iii) Now, variance of X :

$$V(X) = \sigma^2 = E(X^2) - \left[E(X)\right]^2$$

$$= \sum_{x=1}^{5} x^2 \cdot P(X = x) - \left[\mu\right]^2$$

$$= \sum_{x=1}^{5} \left[(1^2)\cdot(0.1) + (2^2)\cdot(0.2) + (3^2)\cdot(0.2) + (4^2)\cdot(0.2) + (5^2)\cdot(0.3)\right] - \left[3.4\right]^2$$ **[1.5 M]**

$$= 13.4 - 11.56$$

$$= 1.84$$

and Standard deviation of X: $\sigma = \sqrt{\sigma^2} = \sqrt{1.84} = 1.357$ **[0.5 M]**

(iv) Now,

$$P(1 < X \le 5) = P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5)$$

$$= 0.2 + 0.2 + 0.2 + 0.3$$ **[1 M]**

$$= 0.9$$

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5. Solve the linear congruence: $24x \equiv 1 \pmod{65}$

Solution:

Step – 1: Using Euclidean Algorithm, we show that $\gcd(24, 65) = 1$.

Now, using Division Algorithm:

$$65 = 24 \cdot 2 + 17 \Rightarrow 65 + 24 \cdot (-2) = 17 \text{ --------} (1)$$
$$24 = 17 \cdot 1 + 7 \Rightarrow 24 + 17 \cdot (-1) = 7 \text{ --------} (2)$$
$$17 = 7 \cdot 2 + 3 \Rightarrow 17 + 7 \cdot (-2) = 3 \text{ --------} (3)$$ **[2 M]**
$$7 = 3 \cdot 2 + 1 \Rightarrow 7 + 3 \cdot (-2) = 1 \text{ --------} (4)$$
$$3 = 1 \cdot 3 + 0$$

Therefore, inverse of 24 under modulo 65 exists, since $\gcd(24, 65) = 1$ (non-zero remainder). **[0.5 M]**

Step – 2: We can express 1 as a linear combination of 24 and 65.

Now

$$\begin{aligned}
(4) &\Rightarrow 7 + 3 \cdot (-2) = 1 \\
&\Rightarrow 7 + [17 + 7 \cdot (-2)] \cdot (-2) = 1, \text{ from (3)} \\
&\Rightarrow 7 \cdot 1 + 17 \cdot (-2) + 7 \cdot 4 = 1 \\
&\Rightarrow 7 \cdot 5 + 17 \cdot (-2) = 1 \\
&\Rightarrow [24 + 17 \cdot (-1)] \cdot 5 + 17 \cdot (-2) = 1, \text{ from (2)} \\
&\Rightarrow 24 \cdot 5 + 17 \cdot (-5) + 17 \cdot (-2) = 1 \\
&\Rightarrow 24 \cdot 5 + 17 \cdot (-7) = 1 \\
&\Rightarrow 24 \cdot 5 + [65 + 24 \cdot (-2)] \cdot (-7) = 1, \text{ from (1)} \\
&\Rightarrow 24 \cdot 5 + 65 \cdot (-7) + 24 \cdot 14 = 1 \\
&\Rightarrow 24 \cdot 19 + 65 \cdot (-7) = 1
\end{aligned}$$

[2 M]

Taking modulo 65 on both sides, we get : $24 \cdot 19 \equiv 1 \pmod{65}$.

Therefore, the solution of the given linear congruence is : $x = 19$. **[0.5 M]**

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